



Faculty of Engineering and Technology
Electrical and Computer Engineering Department
ENCS4130 – Computer Network Laboratory

Experiment No. 03

Dynamic Routing 1 – Distance Vector Routing Protocols (RIP and EIGRP)

■ Objectives

- (a) Learn how to configure and verify IP routing on Cisco routers.
- (b) Understand the principles of dynamic routing using RIP and EIGRP.

■ Requirements

- (a) Cisco Packet Tracer software.
- (b) Three routers.
- (c) Two switches.
- (d) One wireless access point.
- (e) One network sniffer.
- (f) Four PCs, one laptop, one tablet, and one printer.

■ Pre-Lab Task

Students are required to build the network topology shown in Figure 2 and configure the IP addresses, access point, and end devices by completing the steps outlined in Sections 3.4.1 and 3.4.2.

3.1 Introduction

Routing in data communication networks is categorized into *static* and *dynamic* routing. **Static routing** requires manual configuration of routing tables, whereas **dynamic routing** uses protocols to automatically discover networks and update routing tables. While dynamic routing simplifies network management, it consumes additional CPU resources and network bandwidth.

Dynamic routing protocols are classified into **Interior Gateway Protocols (IGPs)** and **Exterior Gateway Protocols (EGPs)**, as illustrated in Figure 1. IGPs manage routing within a single *Autonomous System (AS)*, also known as *intra-AS routing*, while EGPs manage routing between different ASes (*inter-AS routing*). An **AS** is a collection of interconnected IP networks and routers under a single organization's control, presenting a unified routing policy to the Internet.

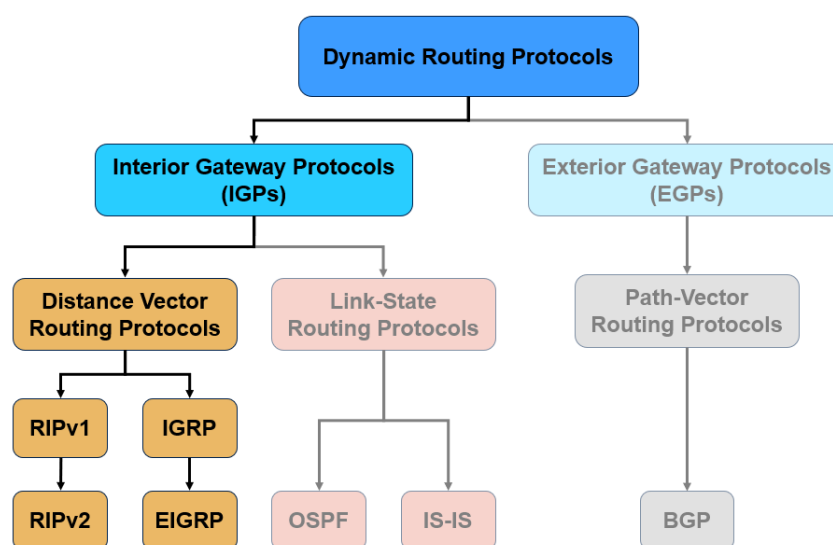


Figure 1: Dynamic routing protocols.

The IGPs are further divided into *distance vector (DV)* routing and *link-state (LS)* routing protocols. This experiment focuses on DV routing protocols, specifically *Routing Information Protocol (RIP)* and *Enhanced Interior Gateway Routing Protocol (EIGRP)*. An example of an LS protocol is *Open Shortest Path First (OSPF)*, which will be covered in the next experiment. The EGPs will be covered in a subsequent experiment.

3.2 Routing Protocol Basics

A **routing protocol** defines the set of rules that routers follow to exchange routing information. In a **dynamic routing protocol**, each router initially knows only the networks directly connected to it. Periodically, routers share (broadcast or multicast) their routing tables—either entirely or as incremental updates—with neighboring routers. Through this iterative exchange, routers gradually learn about all reachable networks and determine the best path to each destination based on next-hop information and distance metrics. When a node or link fails, neighboring routers update their routing tables and propagate the changes throughout the network until all routers converge to a consistent routing view. Before configuring Cisco routers to use RIP and EIGRP, it is essential to understand two key concepts: *administrative distance* and *protocol metrics*.

3.2.1 Administrative Distance (AD)

The **administrative distance** is used to rate the trustworthiness of routing information received by a router from neighboring routers. It is expressed as an *integer between 0 and 255*, where *0* represents the *most trusted* route, and *255* indicates that the route is *completely unreliable* (i.e., the router does not believe the source of that route and does not install it in the routing table).

When a router receives multiple route advertisements for the same network, it first evaluates the AD. The route with the lowest AD is selected and added to the routing table. If multiple routes share the same AD, the router considers protocol metrics (such as hop count or bandwidth) to determine the best path.

Table 1 shows the default AD values used by Cisco routers. By default, a directly connected network is always preferred. If a static route is configured, it takes precedence over dynamically learned routes unless its AD is manually increased. For example, if a router learns a route from both a static configuration and a dynamic protocol (e.g., RIP or EIGRP), it will prioritize the static route unless otherwise specified.

Table 1: The AD for different routing protocols.

Route Source	AD
Connected interface (directly)	0
Static route	1
EIGRP	90
IGRP	100
RIP	120

3.2.2 Metric

A **metric** is a value used by routing protocols to determine the best path to a destination when multiple routes with the same AD exist. A *lower metric value* indicates a *more efficient path*. Routing protocols calculate metrics using various criteria, such as:

- **Hop count** (number of routers a packet must traverse),
- **Bandwidth** (link capacity),
- **Delay** (latency),
- **Reliability** (error rate), and
- **Load** (network congestion).

RIP employs a simple *hop count* metric, selecting the route with the fewest hops. However, RIP has a *maximum hop count of 15*, meaning that any destination beyond 15 hops is considered unreachable. While RIP's simplicity makes it easy to configure, it does not account for link speed or congestion, sometimes leading to suboptimal routing decisions.

On the other hand, **EIGRP** supports up to *255 hops*, with a default limit of 100. Additionally, EIGRP uses a more sophisticated **composite metric**, primarily based on *bandwidth* and *delay*. Unlike RIP, EIGRP dynamically adapts to network changes and

can incorporate additional factors such as *reliability*, *load*, *maximum transmission unit (MTU)* when configured. This results in more efficient route selection, making EIGRP well-suited for large and complex networks.

When multiple routes have the same AD and metric, the router performs **load balancing**, distributing traffic across multiple paths. By default, Cisco routers support up to *four equal-cost paths*, but this can be increased to:

- 6 equal-cost paths for RIP, and
- 32 equal-cost paths for EIGRP.

A major advantage of EIGRP over RIP is its support for *unequal-cost load balancing*, allowing traffic to be distributed proportionally across multiple paths with different metrics. This flexibility enhances network performance by preventing bottlenecks and optimizing bandwidth usage.

3.3 Distance Vector Routing Protocols

Distance Vector (DV) routing protocols determine the best path to a remote network based on distance, measured using protocol-specific metrics. The path with the *lowest* distance is selected as the optimal route. The term “***distance vector***” refers to how routes are advertised—as *vectors consisting of both distance* (a metric representing cost) *and direction* (the next-hop router).

In DV routing, each router maintains a table containing the distances to known networks and periodically shares routing information with its directly connected neighbors. These neighbors integrate the received information into their own routing tables and propagate updates throughout the network. This process, often described as ***routing by rumor***, means that routers trust the information provided by their neighbors rather than independently verifying it.

Two well-known DV protocols are RIP and EIGRP. RIP uses the *Bellman–Ford algorithm* to calculate the best routes. On the other hand, the *Diffusion Update Algorithm (DUAL)* is the engine used by EIGRP for path determination and route maintenance.

3.3.1 Routing Information Protocol (RIP)

The RIP, also known as **RIPv1**, was designed for small network domains that use basic routing without subnetting. In other words, it is a **classful** routing protocol—it assumes default class A, B, or C boundaries and does **not** include subnet mask information in its updates. Additionally, RIPv1 *broadcasts its entire routing table* to all neighboring routers every *30 seconds* using the broadcast address *255.255.255.255*, regardless of whether any changes have occurred.

RIPv2 is an enhanced version of the protocol that addresses several limitations of RIPv1. The primary difference is that RIPv2 is a **classless** routing protocol, meaning it includes subnet mask information in its routing updates. RIPv2 also *supports authentication* between neighboring routers, providing a basic level of security in modern environments.

Like RIPv1, RIPv2 sends *full periodic updates every 30 seconds*, but it can also *send partial, triggered updates* when network changes occur—such as when a link fails or a

new route is added. In addition, RIPv2 transmits routing updates using the *multicast address 224.0.0.9*, instead of the broadcast address used by RIPv1. This improvement reduces unnecessary processing by routers that are not participating in RIP.

To configure RIP on a Cisco router, enable the protocol using the following command:

```
Router(config)# router rip
Router(config-router)# version <version>
```

The `version` command specifies which RIP version the router should use (either **1** or **2**) for sending and receiving routing updates. If the `version` command is omitted, the router defaults to **RIPv1**.

Next, specify which networks should be advertised using:

```
Router(config-router)# network <NETWORK-ID>
```

Here, `NETWORK-ID` represents the network address of the directly connected subnet you want to include in the RIP operation. Each time this command is entered, a single network is added. To advertise multiple networks, repeat the command for each one.

Note:

Although the “**network**” command in both RIPv1 and RIPv2 uses only the *network ID without specifying a subnet mask*, the two versions handle route advertisements differently. **RIPv1** assumes the default class A, B, or C boundaries and does not include subnet mask information in its updates. In contrast, **RIPv2** automatically includes the subnet mask of each interface in its routing updates, supporting Variable Length Subnet Masks (VLSM) and discontinuous networks. Thus, even though the RIPv2 configuration syntax looks classful, its routing behavior is classless.

3.3.2 Enhanced Interior Gateway Routing Protocol (EIGRP)

The EIGRP is an advanced DV routing protocol that offers greater efficiency and faster convergence compared to older protocols such as RIP and IGRP. EIGRP transmits only *incremental (partial) updates*—instead of the entire routing table—to affected neighbors when network changes occur, using the *multicast address 224.0.0.10*. This approach significantly reduces bandwidth consumption and improves overall network performance. Originally developed as a *Cisco-proprietary protocol*, EIGRP has since been released as an *open standard*, enabling wider adoption by other network vendors.

A key distinction between RIP and EIGRP configuration is that EIGRP requires an **AS number**. This AS number is a *16-bit value* ranging from *1 to 65,535* and must be the same on all routers within the same EIGRP domain. Routers that use different AS numbers will not form *neighbor relationships* and therefore cannot exchange routing information. Conceptually, the EIGRP AS number functions as a “process identifier”, defining which EIGRP instance a router belongs to.

To enable EIGRP on a Cisco router, use the following command:

```
Router(config)# router eigrp <AS-NUMBER>
Router(config-router)#
```

Here, `AS-NUMBER` represents the EIGRP autonomous system number, which must match on all routers within the same EIGRP domain.

Similarly to RIPv2, EIGRP supports *classless routing*, but it also allows you to precisely include or exclude specific subnets or interfaces, rather than relying on the router to match by major network boundaries. To advertise networks in EIGRP, use:

```
Router(config-router)# network <NETWORK-ID> [<WILDCARD-MASK>]
```

Here, the `WILDCARD-MASK` specifies the range of addresses to include and is calculated as:

$$\text{Wildcard Mask} = 255.255.255.255 - \text{Subnet Mask}$$

The wildcard mask is *optional*; If omitted, EIGRP assumes a classful network advertisement. As with RIP, the `network` command must be entered separately for each network to be included in the EIGRP process.

3.3.3 Automatic Network Summarization

In most Cisco IOS versions, RIPv2 and EIGRP *enable automatic route summarization by default*. When this feature is active, the protocol summarizes subnet routes to their *classful network boundaries* when advertising them across different major networks. As a result, the router advertises a single classful network address instead of individual subnets.

This behavior can lead to routing issues in networks that use *discontiguous subnets* or *VLSM*. Therefore, it is generally recommended to *disable automatic summarization* using the following command in RIPv2 or EIGRP router configuration mode:

```
Router(config-router)# no auto-summary
```

3.4 Procedure

In this lab, we will connect three routers, two switches, and multiple end devices on different networks. This setup requires configuring routing protocols on the routers to enable communication between the networks. Specifically, we will implement dynamic routing using **RIPv2** and **EIGRP** as the selected protocols.

3.4.1 Building the Topology

Construct the network topology shown in Figure 2, which includes the following devices:

- Routers: Use **Router-PT**.
- Switches: Use **Switch-PT**.
- Access point: From “*Network Devices - Wireless Devices*”, use **AccessPoint-PT**.
- PCs, Laptop, Tablet, and Printer: From “*End Devices*”, use **PC-PT**, **Laptop-PT**, **TabletPC-PT**, and **Printer-PT**.
- Packet Capture: Connect a **Sniffer** between **Router1** and **PC0** to monitor the packets sent by the router.
- Use the “**Automatically Choose Connection Type**” to connect devices.

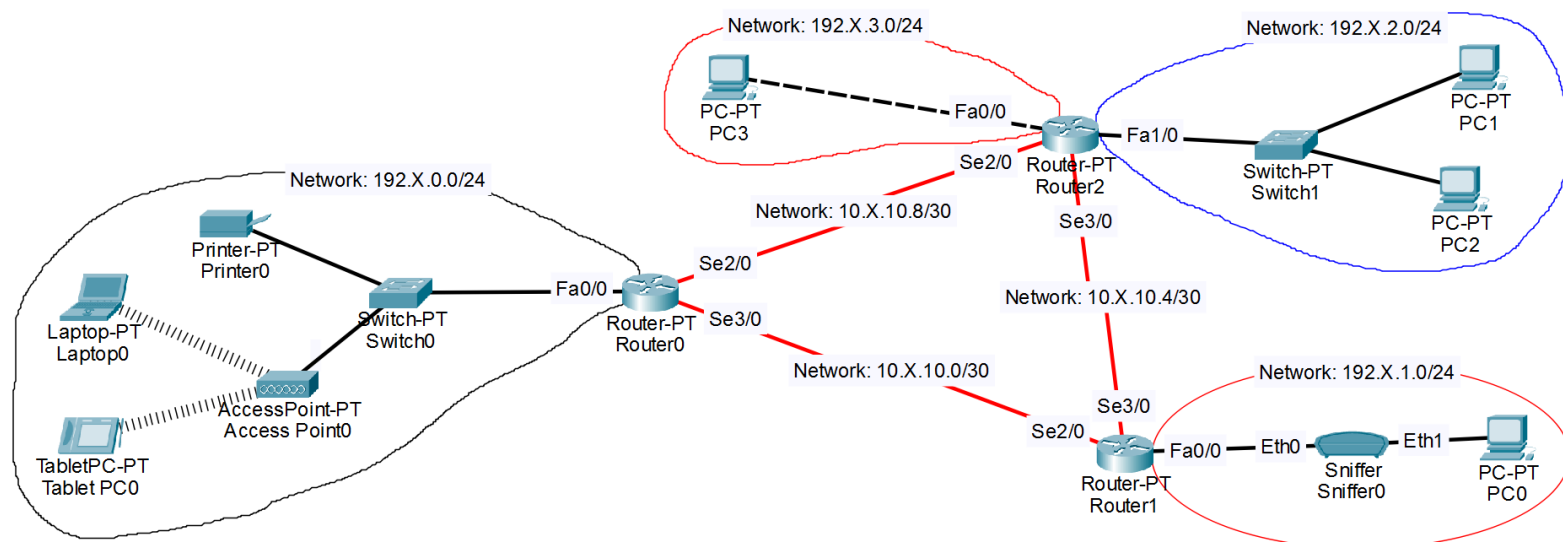


Figure 2: Dynamic routing topology.

3.4.2 Configuring IP Addresses, Access Point, and End Devices

Assign appropriate IP addresses to all end devices and routers, ensuring that each host receives a unique address within the valid host range of its respective network. The value of **X** in the network address corresponds to the *last two digits* of your student ID. For example, if the network address is **192.X.0.0/24** and your student ID is **1230105**, then **X = 05**, resulting in the network address **192.5.0.0/24**.

Notes:

1. The subnet **192.X.0.0/24** includes IP addresses from *192.X.0.0* to *192.X.0.255*, where *192.X.0.0* is the network ID and *192.X.0.255* is the broadcast address.
2. It is recommended to assign the first available host address (e.g., *192.X.0.1*) to router interfaces as gateways.
3. End devices should use addresses from the remaining usable range (e.g., *Printer0: 192.X.0.2*, *Laptop0: 192.X.0.3*, *Tablet PC0: 192.X.0.4*) for proper network configuration.
4. To prevent conflicts, avoid assigning duplicate IP addresses and always verify connectivity after completing the configuration.

A) Configuring the Routers' Interfaces

Set up the interfaces on **Router0** using the following commands:

```
Router(config)# interface FastEthernet0/0
Router(config-if)# ip address 192.X.0.1 255.255.255.0
Router(config-if)# no shutdown
```

```
Router(config)# interface Serial2/0
Router(config-if)# ip address 10.X.10.9 255.255.255.252
Router(config-if)# no shutdown
```

```
Router(config)# interface Serial3/0
Router(config-if)# ip address 10.X.10.1 255.255.255.252
Router(config-if)# no shutdown
```

Configure **Router1** and **Router2** similarly, using their designated IP addresses.

B) Access Point Configuration

The access point provides wireless connectivity within the **192.X.0.0/24** network. To configure the wireless network, go to the “**Config**” tab and set the access point with the following parameters, as shown in Figure 3:

- **Name:** BZU-WiFi
- **Security Model:** WPA2-PSK
- **Pre-Shared Key (PSK):** EnCs4130, wireless devices must enter this key to connect to the access point.
- **Encryption Type:** AES

C) Connecting Wireless Devices

The laptop and tablet wirelessly connect to the **192.X.0.0/24** network through the access point. However, unlike the tablet, the *laptop does not have a built-in wireless*

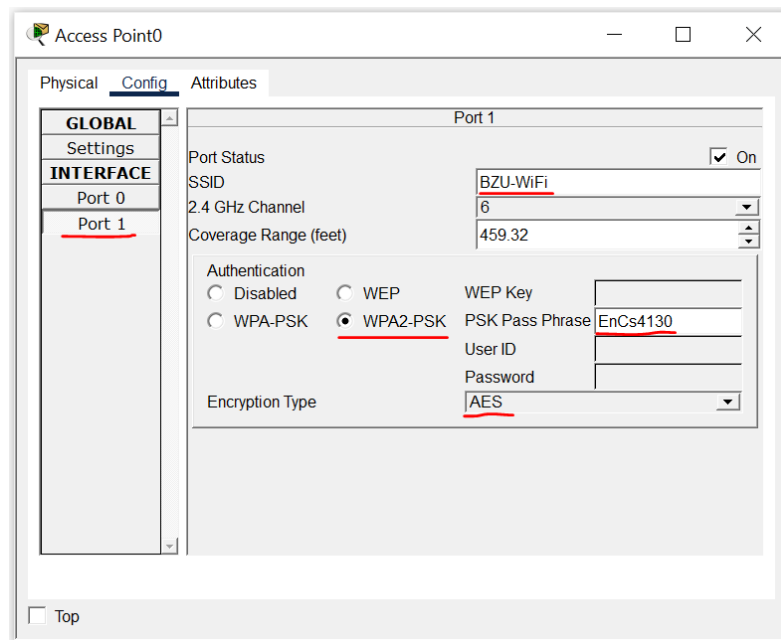


Figure 3: Configuring the access point for Birzeit internal network.

interface. To enable wireless connectivity, you must replace the laptop's wired LAN module with a **WPC300N wireless adapter**.

Follow these steps to install the wireless module on the laptop:

1. Click on the laptop and navigate to the “**Physical**” tab.
2. Turn off the device and remove the existing FastEthernet module.
3. Insert the **WPC300N** wireless adapter from the module list.
4. Power the laptop back on to activate the new module.

Figure 4 illustrates how to connect the laptop to the “BZU-WiFi” network.

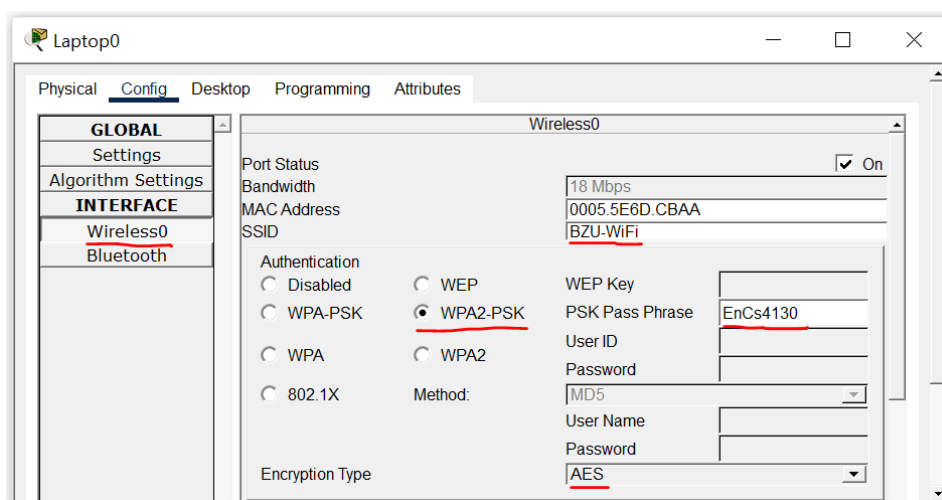


Figure 4: Connecting the laptop to BZU-WiFi.

Repeat the same steps for the **tablet** to connect it to the wireless network.

D) Configuring IP Addresses for End Devices

The end devices are configured with *static IP addresses* and their corresponding *default gateways*. Each device must have a unique IP address within the valid host range of its subnet.

Figures 5 and 6 illustrate the configuration of the Printer and Laptop, respectively.

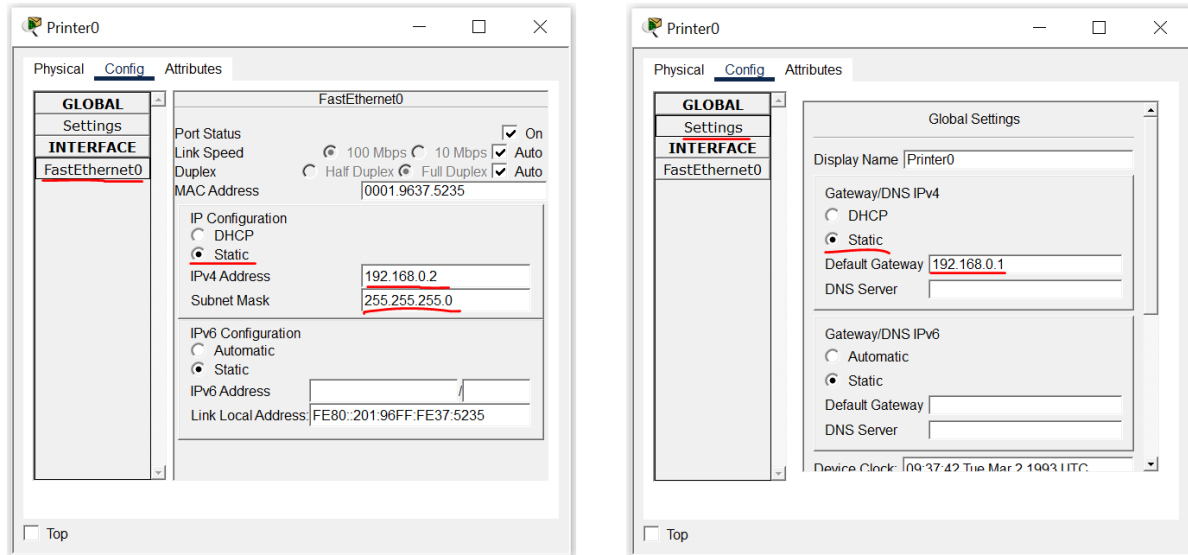


Figure 5: Configuring the Printer.

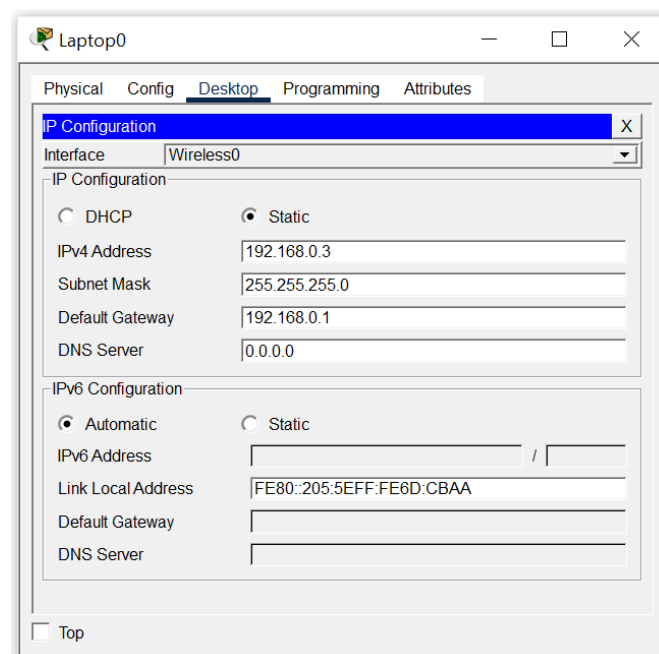


Figure 6: Configuring the Laptop.

Next, configure the **remaining end devices** with their appropriate static IP addresses.

Note:

Before proceeding with the routing configurations, create **two separate copies** of the topology file (.pkt):

- One for **RIPv2** configuration — name it EXP03_RIPv2.pkt
- One for **EIGRP** configuration — name it EXP03_EIGRP.pkt

3.4.3 Configuring RIPv2 Routing (EXP03_RIPv2.pkt Topology File)

RIPv2 advertises only the *directly connected networks* specified in its configuration. Each network that you want RIPv2 to advertise must be *manually added*. Networks that are *not directly connected* will be *learned dynamically* and included in the routing table through RIPv2 updates.

To configure **RIPv2** routing on **Router0**, enter the following commands:

```
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# network 192.X.0.0
Router(config-router)# network 10.X.10.8
Router(config-router)# network 10.X.10.0
Router(config-router)# no auto-summary
```

The `no auto-summary` command disables automatic route summarization, ensuring that subnetted routes are advertised and learned correctly.

As shown, configuring RIPv2 is straightforward and requires only a few commands, making it much simpler than manually defining static routes. However, RIPv2 introduces additional overhead because it periodically exchanges routing updates, which consume CPU processing and network bandwidth.

Now, repeat the configuration steps for **Router1** and **Router2**.

3.4.4 Configuring EIGRP Routing (EXP03_EIGRP.pkt Topology File)

The configuration process for EIGRP is similar to RIPv2, with one key difference: EIGRP requires an AS number. All routers within the same EIGRP domain must use the same AS number to form neighbor relationships and exchange routing information.

For this experiment, you may use any AS number (e.g., AS 1). The following example shows how to enable EIGRP on **Router0** using **AS number 1** and configure the networks accordingly:

```
Router(config)# router eigrp 1
Router(config-router)# network 192.X.0.0 0.0.0.255
Router(config-router)# network 10.X.10.8 0.0.0.3
Router(config-router)# network 10.X.10.0 0.0.0.3
Router(config-router)# no auto-summary
```

The `no auto-summary` command disables EIGRP automatic route summarization, ensuring that subnetted routes are advertised correctly across all routers.

Now, configure EIGRP on **Router1** and **Router2** in the same manner.

3.4.5 Troubleshooting and Verifying Topology Configurations

After configuring the routing protocols, it is essential to verify that all routers and networks can communicate properly. The following commands help you inspect configurations, verify routing behavior, and troubleshoot potential issues.

A) Display Interface Status and IP Addresses

To check the operational status and assigned IP addresses of router interfaces, use:

```
Router# show ip interface brief
```

This command provides a concise summary of all interfaces, including their IP addresses, operational status (up/down), and protocol status.

As shown in Figure 7, **Router2**'s summary confirms that interfaces *FastEthernet0/0*, *FastEthernet1/0*, *Serial2/0*, and *Serial3/0* are enabled, have correct IP configurations, and are fully operational — both the Status and Protocol fields show “up”.

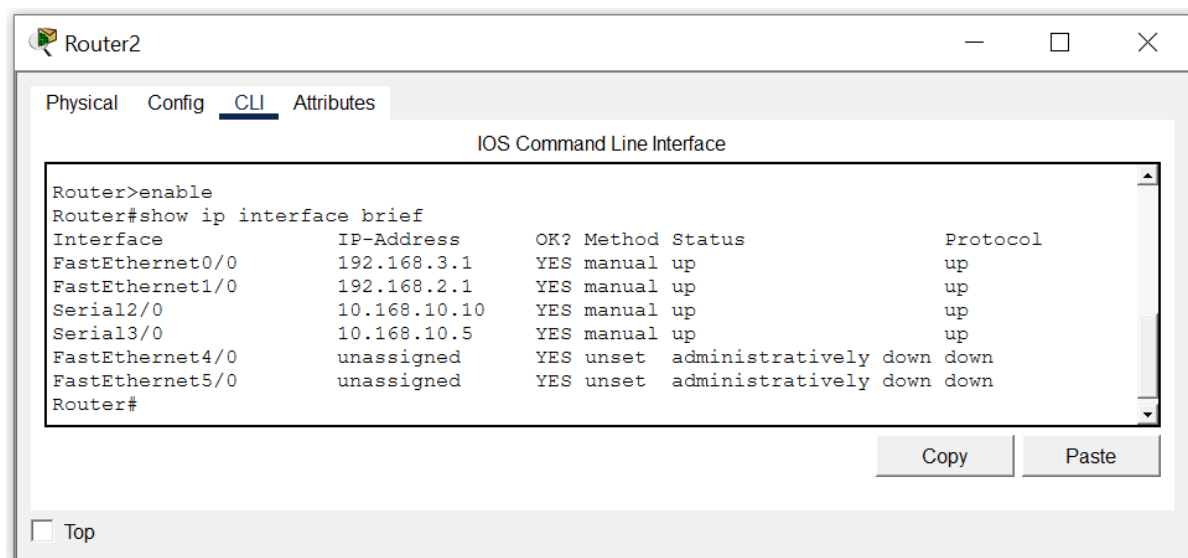


Figure 7: Summary of all interfaces on Router2.

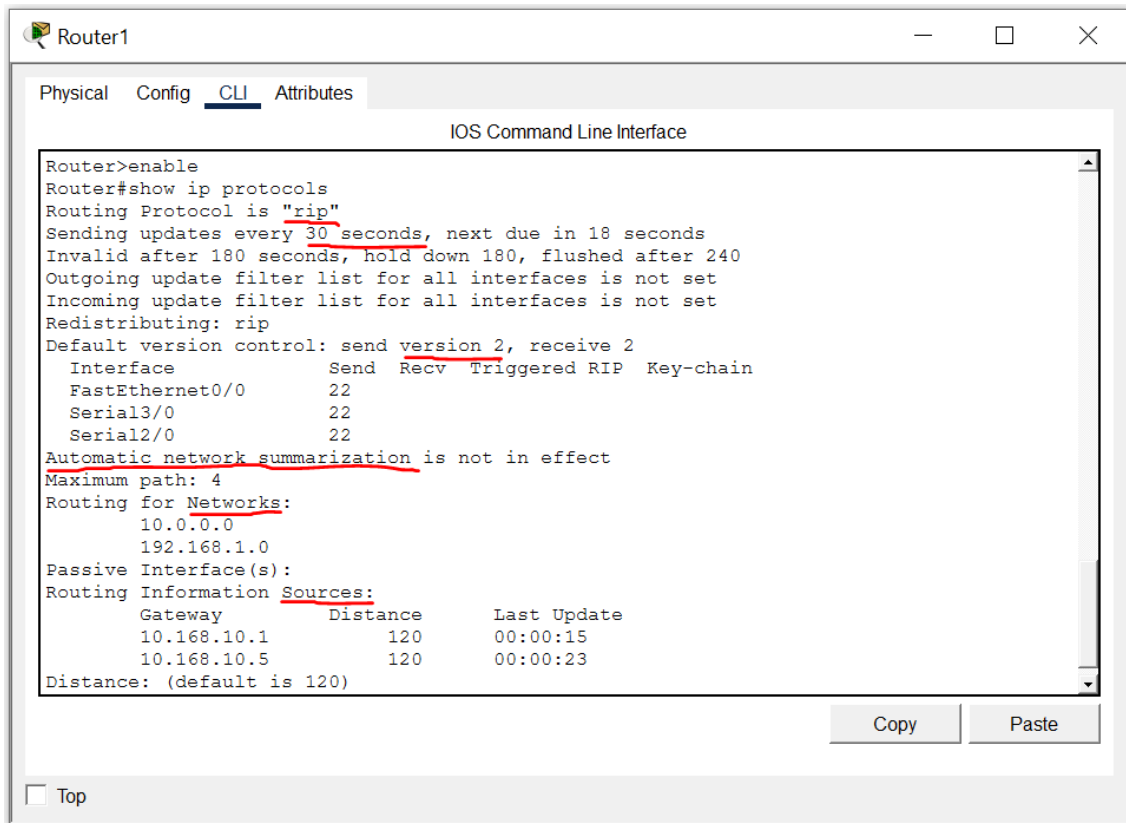
B) Display Active Routing Protocols and Parameters

To verify which dynamic routing protocols are running and review their parameters, use:

```
Router# show ip protocols
```

This command displays the active routing protocols, networks being advertised, update timers, administrative distances, and neighbor relationships.

Figures 8 and 9 show examples of this output for **Router1** running RIPv2 and EIGRP, respectively.



The screenshot shows the Router1 CLI window with the 'CLI' tab selected. The command 'show ip protocols' has been executed, displaying the following output:

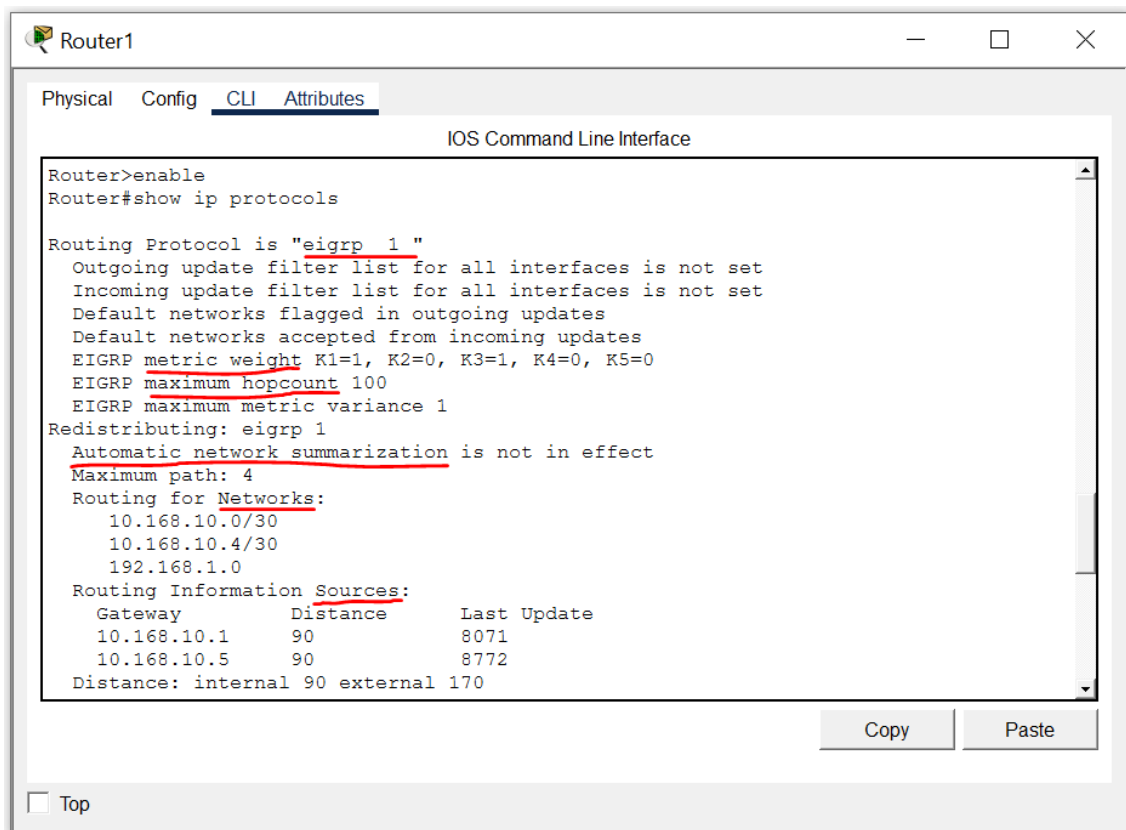
```

Router>enable
Router#show ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 18 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 2, receive 2
  Interface          Send Recv Triggered RIP Key-chain
FastEthernet0/0      22
Serial3/0            22
Serial2/0            22
Automatic network summarization is not in effect
Maximum path: 4
Routing for Networks:
  10.0.0.0
  192.168.1.0
Passive Interface(s):
Routing Information Sources:
  Gateway         Distance      Last Update
  10.168.10.1      120           00:00:15
  10.168.10.5      120           00:00:23
Distance: (default is 120)

```

At the bottom of the window, there are 'Copy' and 'Paste' buttons, and a 'Top' link.

Figure 8: Summary of routing protocols running on Router1 (RIPv2).



The screenshot shows the Router1 CLI window with the 'CLI' tab selected. The command 'show ip protocols' has been executed, displaying the following output:

```

Router>enable
Router#show ip protocols

Routing Protocol is "eigrp 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  EIGRP metric weight K1=1, K2=0, K3=1, K4=0, K5=0
  EIGRP maximum hopcount 100
  EIGRP maximum metric variance 1
Redistributing: eigrp 1
  Automatic network summarization is not in effect
Maximum path: 4
Routing for Networks:
  10.168.10.0/30
  10.168.10.4/30
  192.168.1.0
Routing Information Sources:
  Gateway         Distance      Last Update
  10.168.10.1      90            8071
  10.168.10.5      90            8772
Distance: internal 90 external 170

```

At the bottom of the window, there are 'Copy' and 'Paste' buttons, and a 'Top' link.

Figure 9: Summary of routing protocols running on Router1 (EIGRP).

C) Display the Routing Table

Each router should contain both directly connected routes and dynamically learned routes in its routing table. To view the IPv4 routing table, use:

```
Router# show ip route
```

This command lists all known routes, along with:

- The *routing protocol* used to learn each route.
- The *administrative distance (AD)* and *metric* value.
- The *outgoing interface* to reach each destination.

In Figure 10, the letter “**R**” indicates routes learned dynamically via RIP. The notation [120/1] represents:

- **120** → The **AD** of RIP.
- **1** → The **hop count (RIP metric)** to reach the destination.

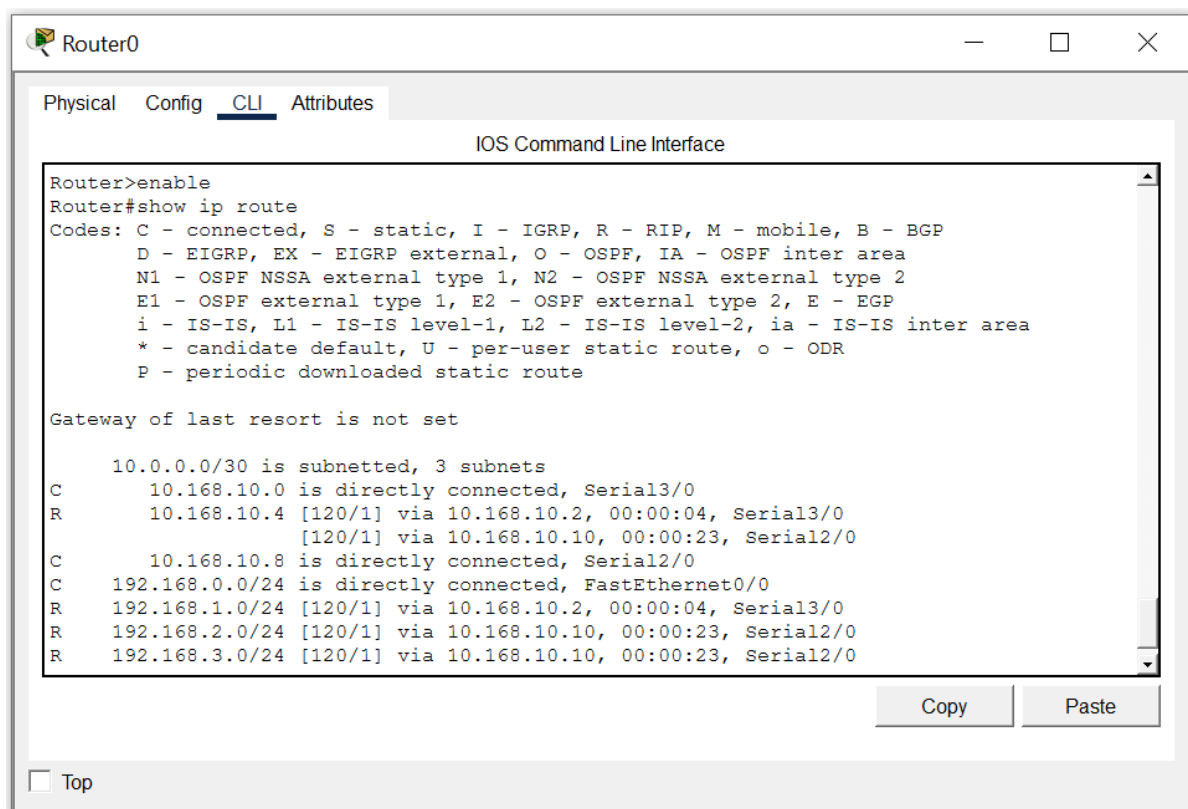


Figure 10: Routing table on Router0 (RIP).

On the other hand, in Figure 11, the letter “**D**” indicates routes learned via EIGRP. The notation [90/2172416] represents:

- **90** → The **AD** of internal EIGRP.
- **2172416** → The **composite EIGRP metric** value to reach the destination.

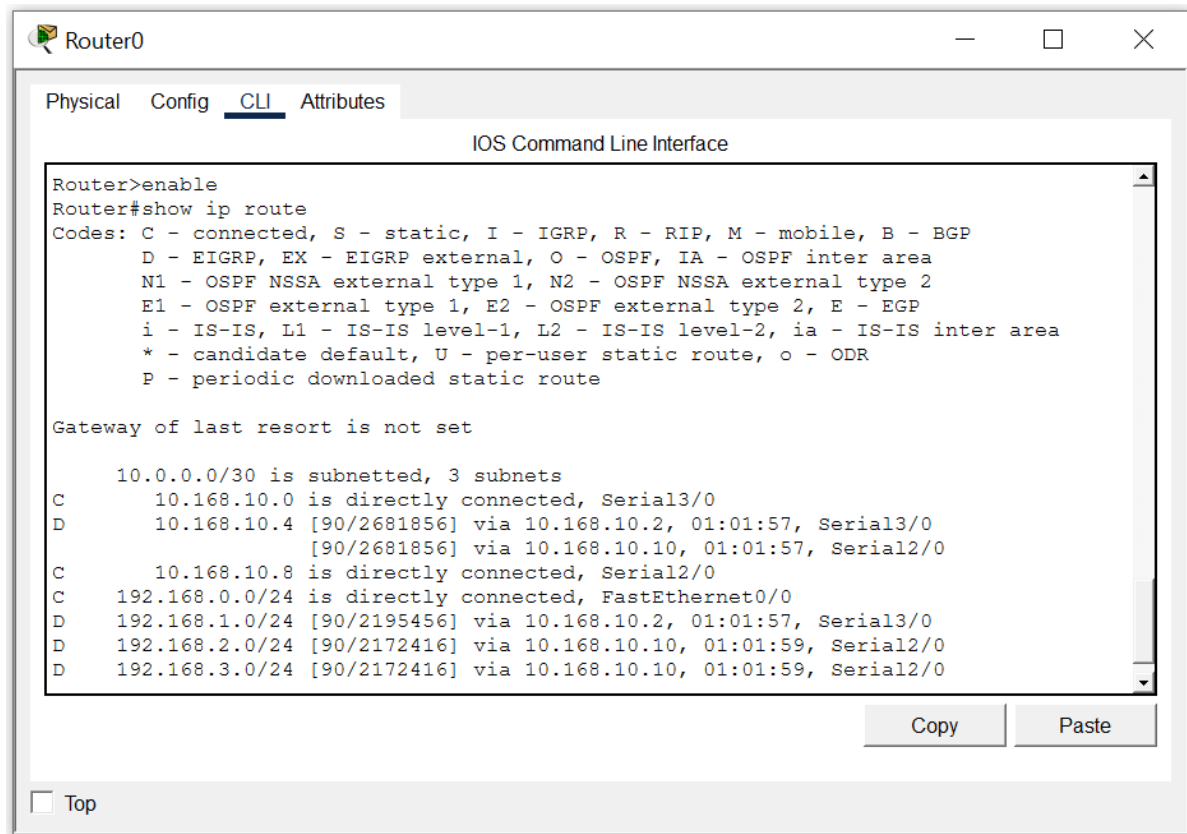


Figure 11: Routing table on Router0 (EIGRP).

Router0 has *two equal-cost paths* to the network **10.X.10.4** — one through **Router1** (*10.X.10.2 via Serial3/0*) and another through **Router2** (*10.X.10.10 via Serial2/0*). Because both paths were learned from the same routing protocol (RIP in Figure 10 or EIGRP in Figure 11) and have *equal metrics* (*1* in RIP, *2681856* in EIGRP), **Router0** installs both routes in its routing table and performs *round-robin load balancing* between them.

D) Monitor EIGRP Packet Exchanges in Real Time

To monitor EIGRP Hello and Update packet activity in real time, use:

```
Router# debug eigrp packets
```

This command displays detailed information about sent and received EIGRP packets, as shown in Figure 12. It is a powerful tool for diagnosing EIGRP communication issues, but should be used with caution, as extensive debugging can affect router performance.

To stop debugging, enter:

```
Router# no debug eigrp packets
```

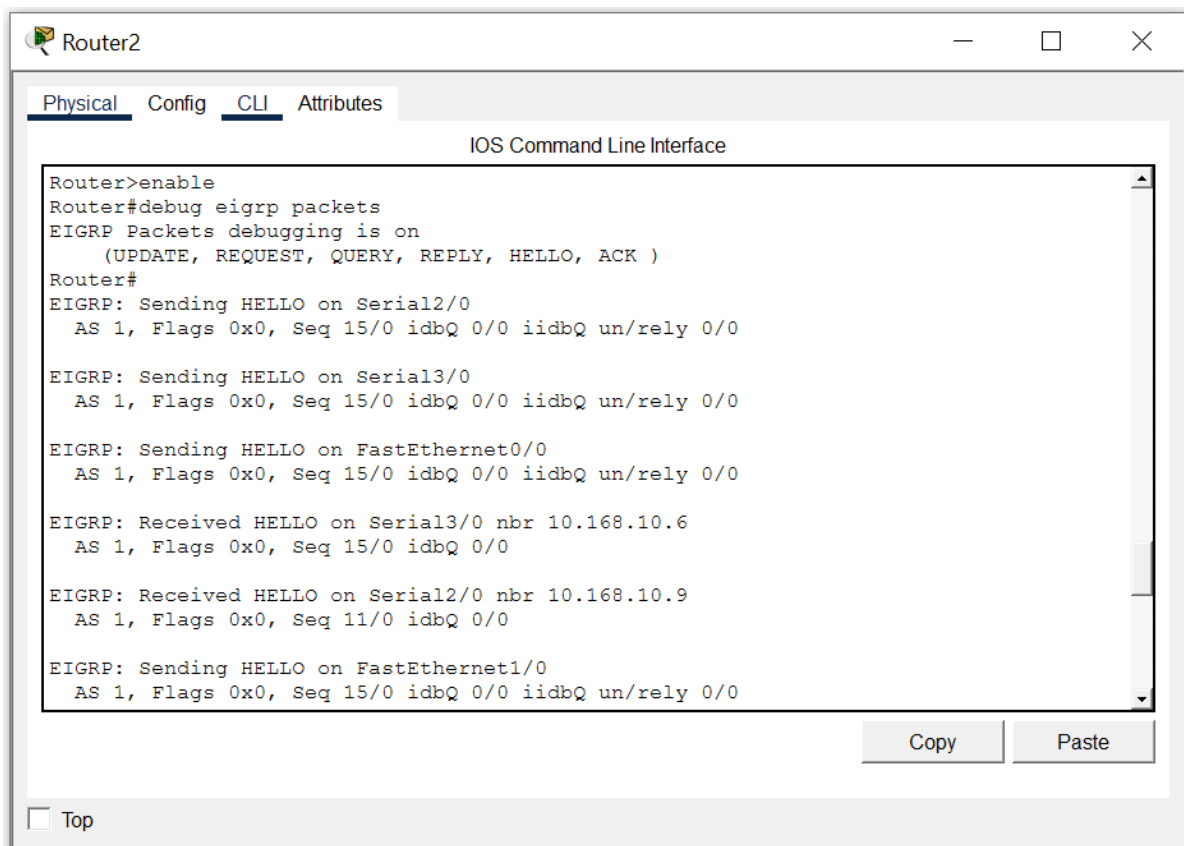


Figure 12: Debugging EIGRP packets on Router2 (EIGRP).

By executing these verification and troubleshooting commands, students can confirm that RIPv2 and EIGRP are correctly configured, routers are successfully exchanging routing information, and the overall network is operating as intended.

3.4.6 Packet Sniffing

To analyze network traffic and understand data flow, the Sniffer tool is used to capture and inspect packets exchanged between devices. Packet sniffing allows students to observe protocol behavior, verify routing operations, and troubleshoot communication issues.

To activate packet capturing, follow these steps:

1. Navigate to **Sniffer** → **GUI** → **Service**.
2. Toggle Check **On/Off** to enable or disable packet capturing.
3. Select **Port0** or **Port1** to specify which interface captures incoming packets. This allows you to control the direction and source of captured traffic.

By default, the sniffer captures all protocols, but students can refine their analysis by applying filters to capture only relevant packets. To clear the current packet capture, click the “**Clear**” button — this removes all previously recorded data from the capture window. For more targeted packet analysis, filters can be applied by selecting “**Edit Filters**” and choosing the desired **protocols** or **packet types** to monitor. For this experiment, focus on the following key protocols:

- **ICMP**: Used for *ping* operations to test connectivity between devices.
- **RIP** and **EIGRP**: Dynamic routing protocols that exchange and update routes.

Applying filters to capture only ICMP, RIP, and EIGRP traffic helps eliminate unnecessary data, making it easier to verify both connectivity and routing behavior. Figure 13 illustrates how to configure and apply these protocol filters. Once the filters are applied, send a packet from **Laptop0** to **PC0**, then inspect the captured data. Clicking on a captured packet reveals detailed information, including: (i) Source MAC address, (ii) Destination MAC address, (iii) Source IP address, and (iv) Destination IP address.

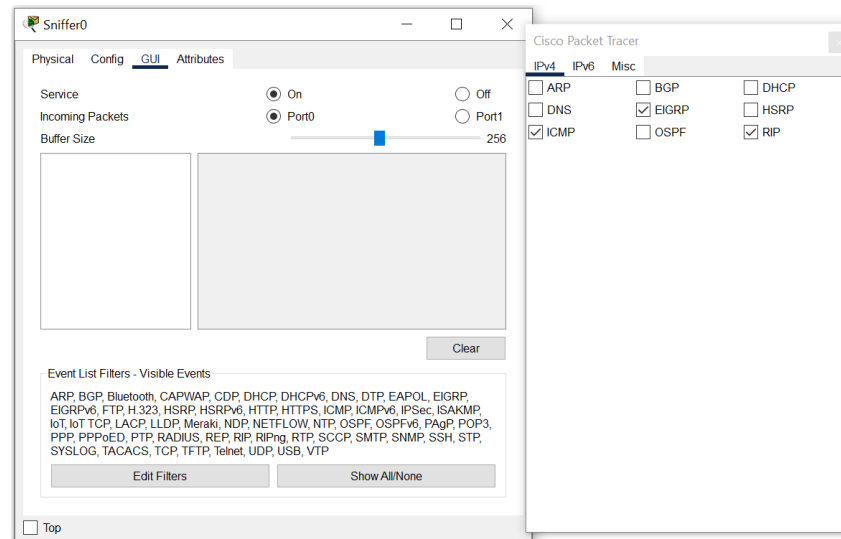


Figure 13: Applying filters to capture specific protocols.

Figure 14 presents the packet details for ICMP traffic, while Figures 15 and 16 illustrate the packet structures for RIP and EIGRP protocols, respectively.

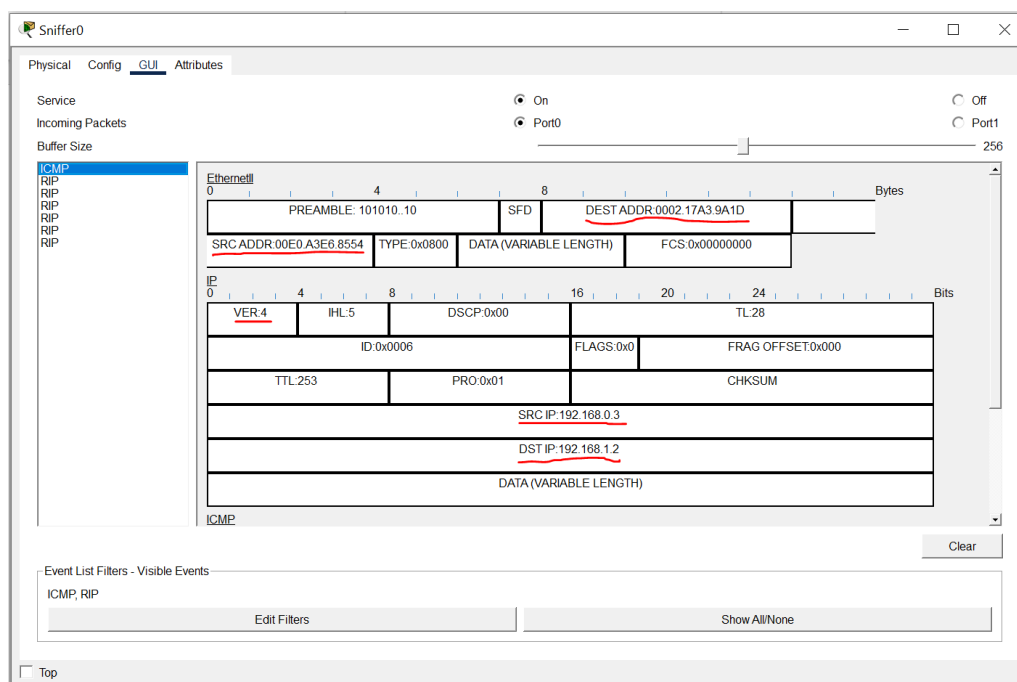


Figure 14: Packet details for ICMP protocol.

Sniffer0

Physical Config **GUI** Attributes

Service ☒ On ☐ Off

Incoming Packets ☒ Port0 ☐ Port1

Buffer Size

RIP

RIP

RIP

Ethernet II

0 4 8 Bytes

PREAMBLE: 101010...10 SFD DEST ADDR: 0100.5E00.0009

SRC ADDR: 00E0.A3E6.8554 TYPE: 0x0800 DATA (VARIABLE LENGTH) FCS: 0x00000000

IP

0 4 8 16 20 24 Bits

VER: 4 IHL: 5 DSCP: 0x00 TL: 152

ID: 0x04c2 FLAGS: 0x0 FRAG OFFSET: 0x000

TTL: 255 PRO: 0x11 CHKSUM

SRC IP: 192.168.1.1

DST IP: 224.0.0.9

DATA (VARIABLE LENGTH)

UDP

0 16 Bits

SOURCE PORT: 520 DESTINATION PORT: 520

LENGTH: 0x0084 CHECKSUM: 0

DATA (VARIABLE LENGTH)

Rip v1

0 8 16 Bits

CMD: 0x02 VER: 0x02 0b0000000000000000

ADDR FAMILY: 0x0 0b0000000000000000

NETWORK

0b0000000000000000

NEXT HOP

METRIC

Clear

Event List Filters - Visible Events

ICMP, RIP

Edit Filters Show All/None

Top

Figure 15: Packet details for RIP protocol.

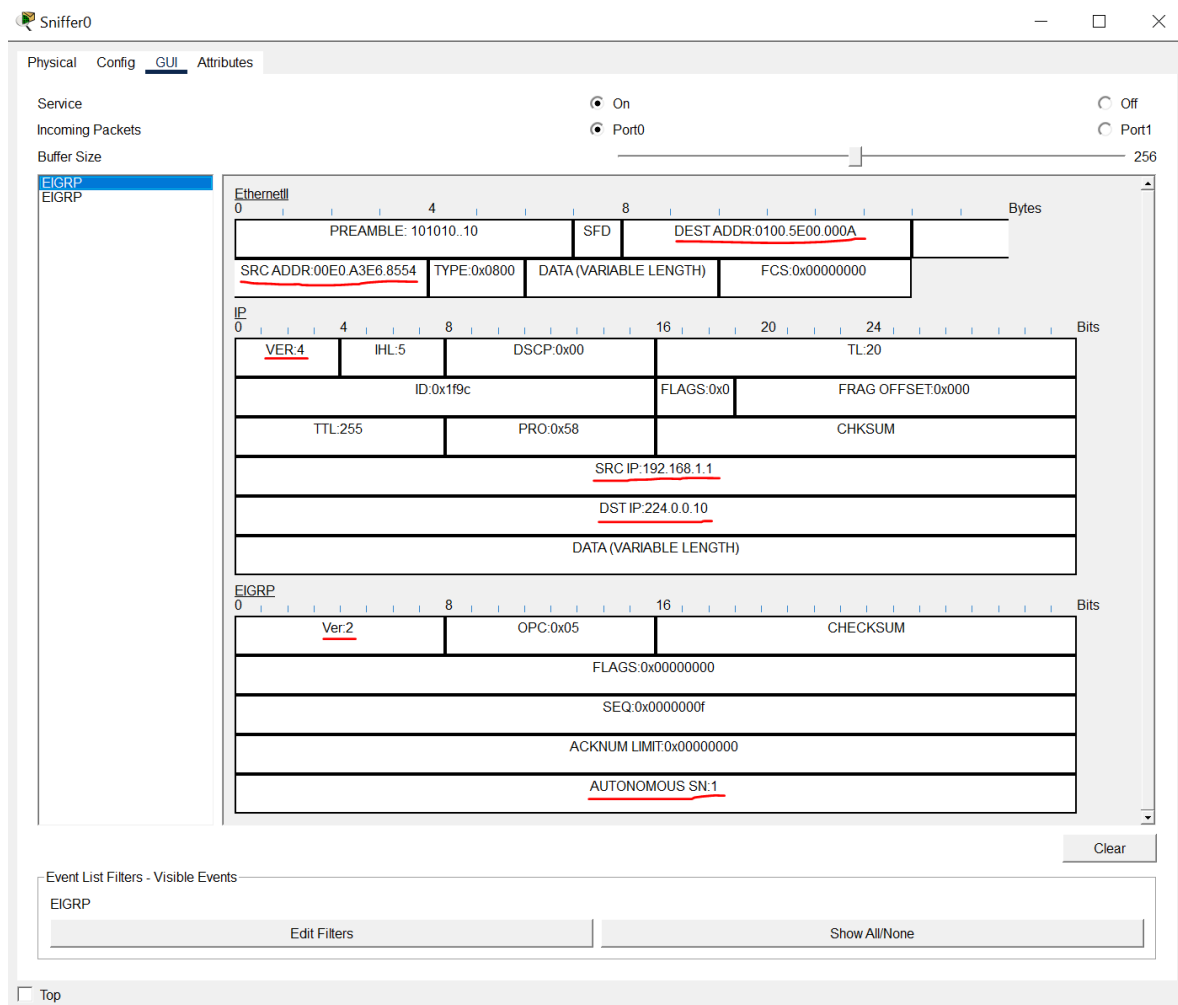


Figure 16: Packet details for EIGRP protocol.

3.5 Simulation Exercise: Link Failure Analysis

In this exercise, you will use **Cisco Packet Tracer's Simulation Mode** to analyze how routing protocols react to link failures. The goal is to observe how routers detect topology changes, exchange routing updates, and adjust their routing tables accordingly.

1. Switch to *Simulation Mode* in Cisco Packet Tracer.
2. Simulate a *link failure* between **Router0** and **Router2** by *shutting down interface Serial2/0* on **Router0** or *Serial3/0* on **Router2**.
3. Observe and capture the packets exchanged between the routers immediately after the link failure occurs.
 - Identify the *type of packets* sent by the routers (e.g., RIP updates or EIGRP update/hello packets).
 - Note the sequence and timing of these control packets.
4. After convergence, display the *updated routing table* on **Router0**.
5. Analyze the changes in the routing table:

- Which routes were removed or updated after the link failure?
 - How did the router determine new paths to the affected networks?
 - What are the metric values of these newly selected routes, and how do they reflect the path cost?
6. *Discuss your observations and findings* with your *instructor or teaching assistant (TA)*. Provide a clear explanation of how the routing protocol (RIPv2 or EIGRP) responded to the topology change and restored network connectivity.

3.6 ToDo

This section will be provided by the instructor.